

On Direction–Magnitude Decomposition in Low-Rank Matrix Optimization

Yudong Wei, Liang Zhang, Bingcong Li, Niao He

Abstract—This work studies direction–magnitude decomposition with a surrogate Riemannian gradient descent method as a framework for low-rank matrix optimization. We apply this framework to overparameterized matrix sensing and completion, and establish linear convergence from random initialization for their population counterpart, namely overparameterized matrix factorization. This yields an exponential acceleration over vanilla gradient descent. Moreover, this framework admits a strictly improved iteration complexity bound compared with standard Riemannian gradient descent. These results highlight the advantages of our proposed framework and indicate its potential for a broader class of low-rank optimization problems.

Index Terms—Direction-magnitude decomposition, Low-rank matrix optimization, Riemannian gradient descent.

I. INTRODUCTION

Low-rank matrices arise in numerous applications across science and engineering, ranging from signal processing [1], [2], machine learning [3], [4], and many others. Consider an optimization problem whose decision variable is a positive semi-definite (PSD) matrix $\mathbf{A} \in \mathbb{S}_+^m$, and suppose that the global optimal is known to be low rank. To exploit this low-rank structure, a widely adopted strategy is the Burer–Monteiro (BM) factorization [5], which represents $\mathbf{A} \in \mathbb{S}_+^m$ as the product of two low-rank factors, namely $\mathbf{A} = \mathbf{Y}\mathbf{Y}^\top$ with $\mathbf{Y} \in \mathbb{R}^{m \times r}$ and $r < m$. This factorization significantly reduces the number of optimization variables and enables scalable algorithms for large-scale problems.

However, the BM factorization does not always guarantee efficient optimization behavior. In particular, overparameterization, which introduces more variables than necessary, is often adopted in practice due to the lack of prior knowledge of the true rank of $\mathbf{A}$. It may lead to optimization difficulties and can even exponentially slow down convergence [6]. One reason for this slowdown comes from that the BM factorization variable $\mathbf{Y}$ entangles the direction information with the magnitude, which can interact unfavorably during optimization.

In this work, we address overparameterized matrix sensing and completion in a unified manner via a direction-magnitude decomposed reformulation and a surrogate Riemannian gradient descent method, referred to as the DMD framework. We establish a linear convergence rate for the corresponding population loss, i.e., overparameterized matrix factorization. This rate is exponentially faster than the sublinear one exhibited by the BM factorization in [6]. Moreover, our analysis yields

a strictly improved iteration complexity bound compared to standard Riemannian gradient descent (RGD) [7].

Our contributions are summarized as follow:

❖ **Exponentially faster convergence.** We propose the DMD framework for overparameterized matrix sensing and completion. At the population level, this framework achieves an exponentially faster convergence rate compared to approaches under the BM factorization.

❖ **Improved condition number dependence.** We establish an iteration complexity bound for DMD that exhibits improved dependence on the condition number κ of the objective matrix $\mathbf{A}$, reducing it from κ^4 to κ^2 , thereby yielding a strictly better complexity guarantee than RGD.

❖ **Empirical validation.** We conduct numerical experiments on overparameterized matrix factorization, and the results corroborate our theoretical findings. We further apply DMD to overparameterized matrix sensing and completion, demonstrating its effectiveness beyond the population setting.

Notational conventions. Bold capital (lowercase) letters denote matrices (column vectors); bold calligraphic letters denote tensors (e.g., $\mathcal{A}$); $\text{Tr}(\cdot)$ denotes the trace of a matrix; $(\cdot)^\top$ and $\|\cdot\|_F$ refer to transpose and Frobenius norm of a matrix; $\|\cdot\|$ denotes the spectral (ℓ_2) norm for matrices (vectors); $\sigma_i(\cdot)$ denotes the i -th largest singular value of a matrix. Moreover, $\mathbb{S}^m$ and $\mathbb{S}_+^m$ denote symmetric and positive semi-definite matrices of size $m \times m$, respectively.

II. PRELIMINARIES

Our goal is to efficiently solve symmetric matrix sensing and matrix completion within a unified framework. Such problems appear in collaborative filtering [8], graph learning [9], and image science [10], among others.

Matrix sensing targets at recovering a ground truth matrix $\mathbf{A} \in \mathbb{S}_+^m$ from a collection of n measured data $\{(\mathbf{M}_i, y_i)\}_{i=1}^n$, where each feature matrix $\mathbf{M}_i \in \mathbb{S}^m$ is symmetric and the corresponding label is $y_i = \text{Tr}(\mathbf{M}_i^\top \mathbf{A})$. Define a linear operator $\mathcal{M} : \mathbb{S}^m \rightarrow \mathbb{R}^n$ with $[\mathcal{M}(\mathbf{A})]_i = \text{Tr}(\mathbf{M}_i^\top \mathbf{A})$. Given that the objective matrix $\mathbf{A}$ is low-rank, i.e., $r_A := \text{rank}(\mathbf{A}) \ll m$, a parameter-efficient formulation can be obtained via the BM factorization. Specifically, a matrix $\mathbf{Y} \in \mathbb{R}^{m \times r}$ is introduced such that $\mathbf{Y}\mathbf{Y}^\top$ serves as an approximation of $\mathbf{A}$, leading to the following problem

$$\min_{\mathbf{Y} \in \mathbb{R}^{m \times r}} \frac{1}{4} \|\mathcal{M}(\mathbf{Y}\mathbf{Y}^\top) - \mathcal{M}(\mathbf{A})\|^2. \quad (1)$$

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Matrix completion aims to recover the ground truth matrix $\mathbf{A} \in \mathbb{S}_+^m$ from a set of observed entries. Let $\mathcal{M}_\Omega : \mathbb{S}^m \rightarrow \mathbb{S}^m$ denote the sampling operator defined by $\mathcal{M}_\Omega(\mathbf{A}) = \Omega \odot \mathbf{A}$, where $\Omega \in \mathbb{S}^m$ is a binary sampling mask and $\odot$ denotes the Hadamard (elementwise) product. Under the BM factorization, the matrix completion problem can be formulated as:

$$\min_{\mathbf{Y} \in \mathbb{R}^{m \times r}} \frac{1}{4} \|\mathcal{M}_\Omega(\mathbf{Y}\mathbf{Y}^\top) - \mathcal{M}_\Omega(\mathbf{A})\|_F^2. \quad (2)$$

To unify the optimization of matrix sensing and completion, and to better understand their intrinsic optimization properties, we study their population loss. Specifically, when the number of measurements tends to infinity and the measurement process is unbiased, the population losses of (1) and (2) coincide and reduce to the following matrix factorization problem:

$$\min_{\mathbf{Y} \in \mathbb{R}^{m \times r}} \frac{1}{4} \|\mathbf{Y}\mathbf{Y}^\top - \mathbf{A}\|_F^2. \quad (3)$$

Despite its simple form, problem (3) is nonconvex with multiple saddle points, which makes convergence to a global optimum from random initialization nontrivial [6], [7], [11]. In practice, overparameterization, i.e., choosing $r > r_A$, is often adopted to ensure exact recovery of $\mathbf{A}$ without prior knowledge of its rank. However, it has been shown in [6] that overparameterization can slow down optimization substantially. In particular, a lower bound for gradient descent (GD) implies that the reconstruction error $\|\mathbf{Y}_t\mathbf{Y}_t^\top - \mathbf{A}\|_F$ cannot converge faster than $\Omega(1/t)$, where t denotes the iteration number. This rate is *exponentially* slower than the linear one achieved in the exactparameterized regime, where the true rank r_A is known and $r = r_A$ is employed [11].

III. DIRECTION-MAGNITUDE DECOMPOSITION

To mitigate the slowdown in convergence induced by overparameterization, next we introduce a direction-magnitude decomposition framework.

A. DMD reformulation

We exploit the polar decomposition [12] on $\mathbf{Y}$, i.e., $\mathbf{Y} = \mathbf{X}\tilde{\Theta}$, where $\mathbf{X} \in \mathbb{R}^{m \times r}$ lies in a Stiefel manifold and $\tilde{\Theta} \in \mathbb{S}_+^r$ is a symmetric positive semi-definite matrix. The Stiefel manifold $\text{St}(m, r)$ is defined as $\{\mathbf{X} \in \mathbb{R}^{m \times r} | \mathbf{X}^\top \mathbf{X} = \mathbf{I}_r\}$. Geometrically, $\mathbf{X}$ is an orthonormal basis spanning an r -dimensional subspace, thereby encoding the directional component of $\mathbf{Y}$, while $\tilde{\Theta}$ captures its magnitude information. Substituting $\mathbf{Y}$ in (3), we arrive at

$$\min_{\mathbf{X} \in \text{St}(m, r), \tilde{\Theta} \in \mathbb{S}_+^r} \frac{1}{4} \|\mathbf{X}\tilde{\Theta}\tilde{\Theta}^\top \mathbf{X}^\top - \mathbf{A}\|_F^2.$$

This problem can be further simplified by absorbing the product $\tilde{\Theta}\tilde{\Theta}^\top$ into a single matrix $\Theta \in \mathbb{S}_+^r$ and relaxing the PSD constraint on Θ to only symmetry, i.e., $\Theta \in \mathbb{S}^r$. Notably, in the overparameterized regime, this relaxation preserves the same global optimum as the original formulation, while substantially improving computational efficiency by avoiding costly operations, such as SVDs or matrix exponentials needed for optimization over PSD cones [13], [14].

Algorithm 1 DMD for solving (4)

- 1: **Require:** Initial point $\mathbf{X}_0 \in \text{St}(m, r)$, stepsizes η, μ
 - 2: **for** $t = 0, 1, \dots, T$ **do**
 - 3: Update the magnitude variable Θ_t via (7)
 - 4: Obtain the surrogate Riemannian gradient $\mathbf{G}_t$ via (5)
 - 5: Update the direction variable $\mathbf{X}_{t+1}$ via (6)
 - 6: **end for**
-

With the proposed direction-magnitude decomposition and the aforementioned simplification, we obtain the following optimization problem:

$$\min_{\mathbf{X} \in \text{St}(m, r), \Theta \in \mathbb{S}^r} f(\mathbf{X}, \Theta) := \frac{1}{4} \|\mathbf{X}\Theta\mathbf{X}^\top - \mathbf{A}\|_F^2. \quad (4)$$

Similar formulations have appeared in [7], [15]. Our work differs by solving (4) in a fully direction–magnitude–decomposed manner, as detailed next.

B. DMD optimization

The key insight behind our algorithm is a separation of optimality in (4). In particular, the optimal $\mathbf{X}$ identifies the eigenspace of $\mathbf{A}$ regardless of whether Θ is optimal. Noticing that this eigenspace can be learned from a closely related problem, principal component analysis (PCA), we use PCA as a surrogate objective for learning $\mathbf{X}$.

Recall the Ky Fan’s characterization of PCA [16]. For a PSD matrix $\mathbf{A} \in \mathbb{S}_+^m$, the leading r -dimensional eigenspace can be obtained by optimizing $\frac{1}{4} \|\mathbf{X}\mathbf{X}^\top - \mathbf{A}\|_F^2$ over $\mathbf{X} \in \text{St}(m, r)$. To handle the manifold constraint on the direction variable $\mathbf{X}$, we optimize it with Riemannian gradient descent. A recap of Riemannian optimization is omitted due to space limitation. We refer the reader to [17], [18] for backgrounds. Denote the $\hat{\mathbf{G}}_t = (\mathbf{X}_t\mathbf{X}_t^\top - \mathbf{A})\mathbf{X}_t$ as the Euclidean gradient (w.r.t. $\mathbf{X}_t$) of the PCA objective, the Riemannian gradient is thus

$$\mathbf{G}_t := (\mathbf{I}_m - \mathbf{X}_t\mathbf{X}_t^\top)\hat{\mathbf{G}}_t + \frac{1}{2}\mathbf{X}_t(\mathbf{X}_t^\top\hat{\mathbf{G}}_t - \hat{\mathbf{G}}_t^\top\mathbf{X}_t). \quad (5)$$

It can be seen that (5) is independent of Θ . In other words, the optimization of the direction is fully decoupled from that of the magnitude. To ensure feasibility after each update, we further apply a polar retraction¹, leading to the following update rule:

$$\mathbf{X}_{t+1} = (\mathbf{X}_t - \eta\mathbf{G}_t)(\mathbf{I}_r + \eta^2\mathbf{G}_t^\top\mathbf{G}_t)^{-1/2}, \quad (6)$$

where $\eta > 0$ denotes the stepsize.

For the magnitude variable Θ , we employ GD on the original objective (4) with stepsize $\mu > 0$, given by

$$\Theta_t = (1 - \frac{\mu}{2})\Theta_{t-1} + \frac{\mu}{2}\mathbf{X}_t^\top\mathbf{A}\mathbf{X}_t. \quad (7)$$

This update preserves the symmetry of Θ_t for all $t \geq 0$ whenever Θ_0 is initialized symmetrically.

In summary, the step-by-step procedure for solving (4) is summarized in Algorithm 1. For convenience, we continue to refer to this algorithm as DMD, since it is fully consistent with the direction-magnitude decomposition paradigm.

¹Let $\mathbf{X} \in \text{St}(m, r)$ and a point in its tangent space $\mathbf{G} \in \mathcal{T}_{\mathbf{X}}\text{St}(m, r)$. The polar retraction for $\mathbf{X} + \mathbf{G}$ is given by $\mathcal{R}_{\mathbf{X}}(\mathbf{G}) = (\mathbf{X} + \mathbf{G})(\mathbf{I}_r + \mathbf{G}^\top\mathbf{G})^{-1/2}$.

TABLE I
COMPARISON WITH EXISTING ALGORITHMS FOR LOW-RANK MATRIX FACTORIZATION. “OP” AND “EP” DENOTE THE OVERPARAMETERIZED AND EXACTPARAMETERIZED SETTINGS, RESPECTIVELY.²

OP / EP	Formulation	Algorithm	Iteration Complexity
EP	(3)	GD [11]	$\mathcal{O}(m^2 r^4 \kappa^4 + m^2 r^4 \kappa^4 \log(\frac{1}{\varepsilon \kappa}))$
OP	(3)	GD [6]	$\mathcal{O}(\frac{\kappa^2}{\log(m r_A^2) \varepsilon})$
OP	(4)	RGD [7]	$\mathcal{O}(\frac{m^4 r^3 r_A \kappa^4}{(r-r_A)^4} + \frac{m^3 r^3 \kappa^4}{(r-r_A)^4} \log(\frac{1}{\varepsilon}))$
OP	(4)	DMD	$\mathcal{O}(\frac{m^2 r^2 r_A^2 \kappa^2}{(r-r_A)^4} + \frac{m r r_A \kappa^2}{(r-r_A)^2} \log(\frac{1}{\varepsilon}))$

IV. BENEFITS OF DMD

We consider DMD under random initialization, where the initial direction variable $\mathbf{X}_0$ is drawn uniformly random from the Stiefel manifold $\text{St}(m, r)$. A practical way to generate such an initialization is to set $\mathbf{X}_0 = \mathbf{Z}_0(\mathbf{Z}_0^\top \mathbf{Z}_0)^{-1/2}$, where the entries of $\mathbf{Z}_0 \in \mathbb{R}^{m \times r}$ are i.i.d. Gaussian random variables $\mathcal{N}(0, 1)$ [19]. Without loss of generality, we assume that $\sigma_1(\mathbf{A}) = 1$ and $\sigma_{r_A}(\mathbf{A}) = 1/\kappa$ with $\kappa > 1$ denoting the condition number. We also denote $r_A := \text{rank}(\mathbf{A})$.

Theorem 1.³ Consider solving the matrix factorization problem (4) initialized with a random $\mathbf{X}_0 \in \text{St}(m, r)$, $r > r_A$. Suppose that $r_A \leq \frac{m}{2}$. Algorithm 1 using stepsizes $\eta = \mathcal{O}(\frac{(r-r_A)^2}{m r r_A \kappa})$ and $\mu = 2$ generates a sequence $\{\mathbf{X}_t, \boldsymbol{\Theta}_t\}_{t=0}^\infty$. With high probability over the initialization, this sequence satisfies that for any $\varepsilon \in (0, 1)$, we have $f(\mathbf{X}_{t_\varepsilon}, \boldsymbol{\Theta}_{t_\varepsilon}) \leq \varepsilon$ after at most $t_\varepsilon = \mathcal{O}(\frac{m^2 r^2 r_A^2 \kappa^2}{(r-r_A)^4} + \frac{m r r_A \kappa^2}{(r-r_A)^2} \log(\frac{1}{\varepsilon}))$ iterations.

This theorem reveals a two-stage convergence behavior of DMD. In the initial phase, the algorithm undergoes a transient period of escaping potential saddle points. After leaving all saddle regions, DMD enters a linear convergence phase, during which the reconstruction error decreases geometrically until exact recovery, i.e., $\lim_{t \rightarrow \infty} \|\mathbf{X}_t \boldsymbol{\Theta}_t \mathbf{X}_t^\top - \mathbf{A}\|_F = 0$. In what follows, we provide a detailed analysis of Theorem 1 to illustrate how the DMD framework enables efficient optimization in overparameterized matrix factorization.

Linear convergence: Theorem 1 shows that DMD converges to the ground-truth matrix $\mathbf{A}$ at a linear rate in the overparameterized setting. In contrast, based on the BM factorization, GD applied to (3) achieves at best a sublinear convergence rate under overparameterization [6].

Improved κ dependence: In terms of the dependence on the condition number κ , DMD improves the scaling from κ^4 , which arises for GD in the exactparameterized regime and for RGD in the overparameterized regime, to κ^2 . This reduced κ -dependence indicates that the proposed DMD is particularly suitable for ill-conditioned problems. Moreover, since $r - r_A <$

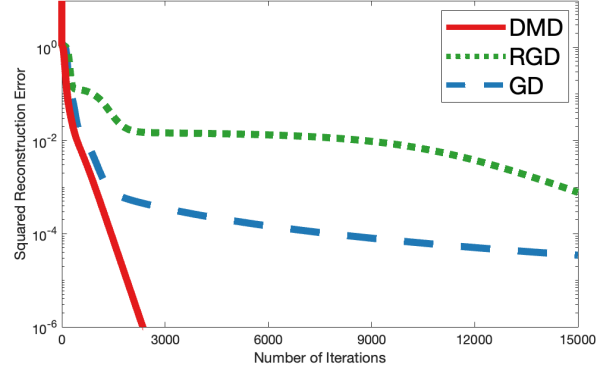


Fig. 1. Convergence comparison of DMD with RGD and GD on overparameterized matrix factorization (squared reconstruction error vs. iteration).

$r < m$, our iteration complexity bound strictly improves upon those of both GD and RGD.

A detailed comparison of iteration complexity bounds is summarized in Table I.

V. NUMERICAL EXPERIMENTS

In this section, we present numerical experiments that validate the theoretical results for the proposed DMD framework in overparameterized matrix problems.

In all subsequent experiments, $\mathbf{A} \in \mathbb{R}^{m \times m}$ is constructed as $\mathbf{A} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{U}^\top$, where $\mathbf{U} \in \mathbb{R}^{m \times r_A}$ has orthonormal columns drawn at random and $\boldsymbol{\Sigma} \in \mathbb{S}_+^{r_A}$ is a diagonal matrix whose entries are evenly spaced on a logarithmic scale within the interval $[1/\kappa, 1]$. The initialization is set as $\mathbf{X}_0(\mathbf{Y}_0) = \mathbf{Z}_0(\mathbf{Z}_0^\top \mathbf{Z}_0)^{-1/2}$, where $\mathbf{Z}_0 \in \mathbb{R}^{m \times r}$ has i.i.d. standard Gaussian entries. The remaining parameters, including m, r_A, r, κ and stepsizes η, μ are specified individually in the description of each experiment.

A. Matrix factorization

We compare DMD against GD and RGD on an overparameterized matrix factorization instance with $m = 100, r_A = 3, r = 20$, and $\kappa = 8$. DMD and RGD are applied to problem (4) with stepsizes $\eta = 0.02$ and $\mu = 2$, and GD is run on problem (3) with a stepsize of $\eta = 0.02$.

As shown in Figure 1, DMD converges faster than both GD and RGD in the initial phase, which is consistent with the weaker κ -dependence of our iteration complexity analysis. After this stage, DMD enters a linear convergence regime and continues to decrease the error until exact recovery of the true matrix is achieved. In contrast, RGD requires a substantially longer time to escape saddle points, subsequently exhibits a slower linear convergence rate than DMD, in agreement with strictly improved iteration complexity of DMD relative to RGD. Moreover, GD quickly slows down to a sublinear rate after the initial phase, resulting in significantly larger reconstruction errors at the same iteration count.

²DMD focuses on symmetric objective matrices, whereas the other algorithms in the comparison are developed for asymmetric objective matrices.

³The proof is provided in the Appendix of the supplementary material, available at [this link](#).

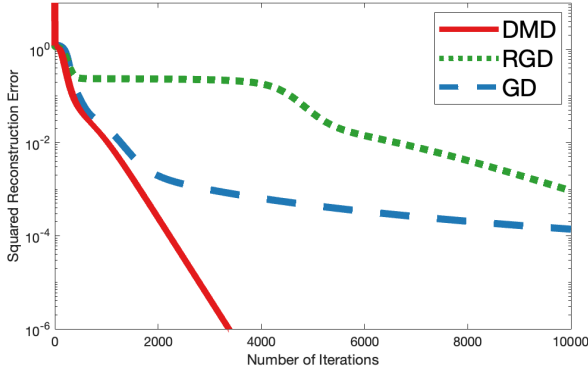


Fig. 2. Convergence comparison of DMD with RGD and GD on overparameterized matrix sensing (squared reconstruction error vs. iteration).

B. Matrix sensing

We further apply DMD to an overparameterized matrix sensing problem to demonstrate the effectiveness of the proposed framework on overparameterized matrix sensing, and compare its performance with RGD and GD.

In this experiment, we consider an instance of $m = 50, r_A = 3, r = 10$, and $\kappa = 5$ with $n = 8000$ sensing matrices generated as $\mathbf{M}_i = \frac{1}{2\sqrt{n}}(\mathbf{R}_i + \mathbf{R}_i^\top)$, $i = 1, \dots, n$, where $\mathbf{R}_i \in \mathbb{R}^{m \times m}$ has i.i.d. standard Gaussian entries. We run DMD and RGD with stepsizes $\eta = 0.01$ and $\mu = 2$ and GD with a stepsize of $\eta = 0.01$.

As we can see in Figure 2, DMD exhibits a linear convergence rate after a short initial transient. In contrast, RGD transitions much more slowly out of its initial stage and converges linearly at a slower rate than DMD. Meanwhile, GD exhibits only sublinear convergence.

C. Matrix completion

Finally, we conduct an experiment on overparameterized matrix completion. We set $m = 100, r_A = 3, r = 25$, and $\kappa = 5$ in the experiment. The sampling mask is generated as $\Omega = \frac{1}{2}(\Xi + \Xi^\top)$, where diagonal entries of $\Xi \in \mathbb{R}^{m \times m}$ are equal to 1 and each off-diagonal entry is independently set to 1 with probability $p = 0.8$. The stepsizes of DMD and RGD are set as $\eta = 0.01, \mu = 2$ and the stepsize of GD is $\eta = 0.01$.

From Figure 3, we observe that DMD retains its linear convergence behavior in the overparameterized matrix completion setting, which is exponentially faster than that of GD. Moreover, the onset of the linear convergence regime for DMD occurs substantially earlier than that for RGD and DMD converges at a faster linear rate than RGD.

VI. FUTURE EXTENSIONS

A. Extensions to asymmetric problems

Throughout this paper, we focus on symmetric objective matrices $\mathbf{A}$. Nevertheless, the proposed direction-magnitude decomposition framework naturally extends to asymmetric settings. Consider a general matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ with BM factorization $\mathbf{Y}_1 \mathbf{Y}_2^\top$, where $\mathbf{Y}_1 \in \mathbb{R}^{m \times r}$ and $\mathbf{Y}_2 \in \mathbb{R}^{n \times r}$.

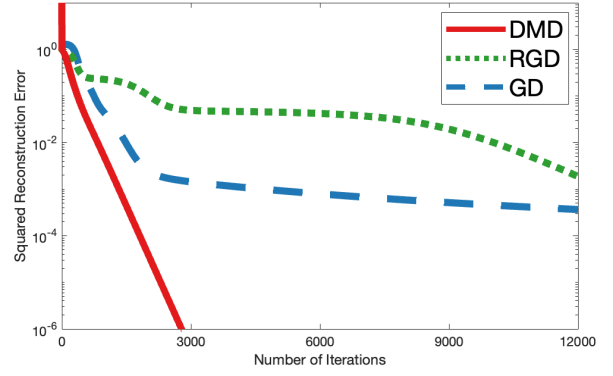


Fig. 3. Convergence comparison of DMD with RGD and GD on overparameterized matrix completion (squared reconstruction error vs. iteration).

Applying DMD yields $\mathbf{Y}_1 = \mathbf{X}_1 \Theta_1$, $\mathbf{Y}_2 = \mathbf{X}_2 \Theta_2$, with $\mathbf{X}_1 \in \text{St}(m, r)$, $\mathbf{X}_2 \in \text{St}(n, r)$ and $\Theta_1, \Theta_2 \in \mathbb{S}_+^r$. By combining the two magnitude matrices into $\Theta = \Theta_1 \Theta_2^\top \in \mathbb{R}^{r \times r}$, we obtain the representation $\mathbf{A} = \mathbf{X}_1 \Theta \mathbf{X}_2^\top$. This formulation shows that DMD provides a unified framework for both symmetric and asymmetric low-rank matrix optimization problems.

B. Extensions to tensor problems

Low-rank tensor optimization problems have attracted increasing attention. For instance, [20] develops a scale-invariant relaxation for recovering tensors from incomplete measurements, and [21] tackles tensor robust principal component analysis under the Tucker decomposition. The Tucker decomposition [22] can be viewed as a natural extension of the DMD to tensor optimization problems. Specifically, for a d -th order tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times \dots \times n_d}$ with multilinear rank $(r_1, \dots, r_d)$, the Tucker decomposition represents $\mathcal{A}$ as $\mathcal{A} = (\mathbf{X}_1, \dots, \mathbf{X}_d) \cdot \mathcal{C}$, where $\mathbf{X}_k \in \text{St}(n_k, r_k)$ for $k = 1, \dots, d$ are the direction variables associated with each mode, and $\mathcal{C} \in \mathbb{R}^{r_1 \times \dots \times r_d}$ is the core tensor capturing the magnitude information. This decomposition directly parallels the DMD in the matrix setting, with the Stiefel factors encoding directional subspaces and the core tensor aggregating the magnitude. Consequently, we expect that the DMD framework can be extended to tensor optimization problems with only minor modifications.

VII. CONCLUSION

This work provides a general direction-magnitude decomposition (DMD) framework for low-rank matrix optimization. We establish that DMD achieves a linear convergence rate for overparameterized matrix factorization, yielding an exponential acceleration over vanilla gradient descent under the BM factorization. Moreover, the proposed framework admits a strictly improved iteration complexity bound compared to standard Riemannian gradient descent. Numerical experiments corroborate our theoretical findings and further demonstrate the effectiveness of DMD on overparameterized matrix sensing and matrix completion. An interesting direction for future work is to rigorously characterize the benefits of the DMD framework in tensor optimization problems.

REFERENCES

- [1] E. J. Candes, T. Strohmer, and V. Voroninski, “Phaselift: Exact and stable signal recovery from magnitude measurements via convex programming,” *Communications on Pure and Applied Mathematics*, vol. 66, no. 8, pp. 1241–1274, 2013.
- [2] Z.-Q. Luo, W.-K. Ma, A. M.-C. So, Y. Ye, and S. Zhang, “Semidefinite relaxation of quadratic optimization problems,” *IEEE Signal Processing Magazine*, vol. 27, no. 3, pp. 20–34, 2010.
- [3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen *et al.*, “Lora: Low-rank adaptation of large language models,” *ICLR*, vol. 1, no. 2, p. 3, 2022.
- [4] B. Li, L. Zhang, A. Mokhtari, and N. He, “On the crucial role of initialization for matrix factorization,” *arXiv preprint arXiv:2410.18965*, 2024.
- [5] S. Burer and R. D. Monteiro, “A nonlinear programming algorithm for solving semidefinite programs via low-rank factorization,” *Mathematical programming*, vol. 95, no. 2, pp. 329–357, 2003.
- [6] N. Xiong, L. Ding, and S. S. Du, “How over-parameterization slows down gradient descent in matrix sensing: The curses of symmetry and initialization,” in *International Conference on Learning Representations*, 2024.
- [7] K. Lion, L. Zhang, B. Li, and N. He, “Polar: Polar-decomposed low-rank adapter representation,” *arXiv preprint arXiv:2506.03133*, 2025.
- [8] J. B. Schafer, D. Frankowski, J. Herlocker, and S. Sen, “Collaborative filtering recommender systems,” in *The adaptive web: methods and strategies of web personalization*. Springer, 2007, pp. 291–324.
- [9] F. Xia, K. Sun, S. Yu, A. Aziz, L. Wan, S. Pan, and H. Liu, “Graph learning: A survey,” *IEEE Transactions on Artificial Intelligence*, vol. 2, no. 2, pp. 109–127, 2021.
- [10] R. Horstmeyer, R. Y. Chen, X. Ou, B. Ames, J. A. Tropp, and C. Yang, “Solving ptychography with a convex relaxation,” *New journal of physics*, vol. 17, no. 5, p. 053044, 2015.
- [11] T. Ye and S. S. Du, “Global convergence of gradient descent for asymmetric low-rank matrix factorization,” *Advances in Neural Information Processing Systems*, vol. 34, pp. 1429–1439, 2021.
- [12] N. J. Higham, “Computing the polar decomposition—with applications,” *SIAM Journal on Scientific and Statistical Computing*, vol. 7, no. 4, pp. 1160–1174, 1986.
- [13] L. Vandenberghe and S. Boyd, “Semidefinite programming,” *SIAM review*, vol. 38, no. 1, pp. 49–95, 1996.
- [14] M. J. Todd, “Semidefinite optimization,” *Acta Numerica*, vol. 10, pp. 515–560, 2001.
- [15] Y. Wei, L. Zhang, B. Li, and N. He, “On the benefits of weight normalization for overparameterized matrix sensing,” *arXiv preprint arXiv:2510.01175*, 2025.
- [16] K. Fan, “On a theorem of weyl concerning eigenvalues of linear transformations i,” *Proceedings of the National Academy of Sciences*, vol. 35, no. 11, pp. 652–655, 1949.
- [17] N. Boumal, *An introduction to optimization on smooth manifolds*. Cambridge University Press, 2023.
- [18] P.-A. Absil, R. Mahony, and R. Sepulchre, “Optimization algorithms on matrix manifolds,” in *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2009.
- [19] Y. Chikuse, *Statistics on Special Manifolds*. Springer Science & Business Media, 2012, vol. 174.
- [20] H. Zheng, Y. Lou, G. Tian, and C. Wang, “A scale-invariant relaxation in low-rank tensor recovery with an application to tensor completion,” *SIAM Journal on Imaging Sciences*, vol. 17, no. 1, pp. 756–783, 2024.
- [21] H. Dong, T. Tong, C. Ma, and Y. Chi, “Fast and provable tensor robust principal component analysis via scaled gradient descent,” *Information and Inference: A Journal of the IMA*, vol. 12, no. 3, pp. 1716–1758, 2023.
- [22] L. R. Tucker, “Some mathematical notes on three-mode factor analysis,” *Psychometrika*, vol. 31, no. 3, pp. 279–311, 1966.